

# Israeli Tick Typhus (Israeli Spotted Fever): A Comprehensive Disease Characteristics Report

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**Disease:** Israeli Tick Typhus / Israeli Spotted Fever (ISF) **MONDO ID:** MONDO:0000230  
**Category:** Infectious Disease (tick-borne spotted-fever-group rickettsiosis) **Causative agent:** *Rickettsia conorii* subsp. *israelensis* **Identifiers:** MONDO:0000230 · ICD-10 A77.1 · ICD-11 1C30.0 · MeSH D001907 (Boutonneuse Fever)

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## Summary

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Israeli Tick Typhus, more precisely termed **Israeli Spotted Fever (ISF)**, is a tick-borne, spotted-fever-group (SFG) rickettsiosis caused by the obligate intracellular Gram-negative bacterium *Rickettsia conorii* subsp. *\*israelensis*. *It is a clinical and etiological variant of Mediterranean spotted fever (MSF) / boutonneuse fever, distinguished from the classic R. conorii subsp. conorii (Malish) strain by multi-locus sequence typing (MLST) and by a more virulent clinical phenotype. The disease is transmitted by the brown dog tick, Rhipicephalus sanguineus\*\*\*, which serves as both vector and reservoir, with domestic dogs acting as amplifying/sentinel hosts. This report synthesizes 16 confirmed findings drawn from 61 papers reviewed across a five-iteration investigation.*

The core pathophysiology is a **disseminated small-vessel vasculitis**: rickettsiae enter through the dermis at the tick-bite site, spread hematogenously, and infect **vascular endothelial cells**—most critically in the brain and lungs—driving increased microvascular permeability, edema, platelet/coagulation activation, and multi-organ dysfunction. Clinically, fever and maculopapular rash are near-universal, but unlike classic MSF the diagnostic eschar (*tache noire*) is frequently **absent**, and a history of tick bite is often lacking, which contributes to diagnostic delay. The ISF strain is associated with a **case-fatality of ~20–30%** in hospitalized/severe cohorts, with older age, alcoholism, G6PD deficiency, and delayed treatment as major prognostic risk factors.

Management is straightforward and highly effective when timely: **doxycycline** is the drug of choice and is curative even as a single-day 200 mg course, with josamycin, clarithromycin, and (historically) chloramphenicol as alternatives for children and pregnant women. There is **no vaccine**; prevention rests on tick-bite avoidance and integrated *Rhipicephalus sanguineus* vector control on dogs and premises. ISF is a **zoonosis** whose geographic range—historically the Mediterranean basin (Israel, Italy, Portugal)—is now documented across multiple continents (Iran, Uganda, Ghana), reflecting the global spread of the brown dog tick.

## Key Findings

### Finding 1 — Disease identity: ISF is caused by *R. conorii* subsp. *israelensis*, one of four MLST-defined subspecies

Multi-locus sequence typing of 39 isolates/tick amplicons resolved four *R. conorii* genotypes: the **Malish** type, **Indian tick typhus (ITTR)** type, **Astrakhan fever (AFR)** type, and the **Israeli spotted fever (ISFR)** type. Pairwise similarity across the *16S rRNA*, *gltA*, *ompA*, *ompB*, and *sca4* loci ranged 98.2–100%, supporting classification at the **subspecies (not species) rank** (PMID: 15766388). ISF is therefore *R. conorii* subsp. *israelensis*, a member of the *R. conorii* complex and a variant of Mediterranean spotted fever / boutonneuse fever (PMID: 28408252).

"Among the 39 isolates or tick amplicons studied, four MLST genotypes were identified: i) the Malish type; ii) the ITTR type; iii) the AFR type; and iv) the ISFR type." — PMID: 15766388

**Key identifiers:** MONDO:0000230; **ICD-10 A77.1** (Spotted fever due to *Rickettsia conorii*; includes Boutonneuse/Marseilles/Mediterranean tick fever); **ICD-11 1C30.0**; **MeSH D001907** (Boutonneuse Fever); NCBI Taxonomy pathogen *Rickettsia conorii* subsp. *israelensis*. **Synonyms:** Israeli spotted fever, Israeli tick typhus, a form of Mediterranean spotted fever / boutonneuse fever / fièvre boutonneuse. Information is derived from **aggregated disease-level sources** (case series, cohorts, case reports), not single-patient EHR data.

## Finding 2 — The ISF strain is more virulent than Malish; alcoholism and older age predict fatal outcome

A prospective Portuguese study of **140 strain-identified *R. conorii* patients (1994–2006; 71 Malish, 69 ISF)** found that **29 adults (21%) died**, and that fatal outcome was **significantly more likely with the ISF strain**, with **alcoholism** identified as a risk factor ([PMID: 18582199](#)). The pathophysiology of fatal disease involved significantly greater incidence of petechial rash, gastrointestinal symptoms, obtundation/confusion, dehydration, tachypnea, hepatomegaly, leukocytosis, coagulopathy, azotemia, hyperbilirubinemia, and elevated hepatic enzymes and creatine kinase.

*"A fatal outcome was significantly more likely for patients infected with the ISF strain, and alcoholism was a risk factor." — [PMID: 18582199](#)*

Note a nuance: a separate Portuguese analysis (n=94) found that neither eschar presence (Malish 49% vs ISF 39%) nor fatality differed *statistically* between strains ([PMID: 16481514](#)), suggesting the "greater severity" concept, while supported by the larger prospective cohort and Israeli national data, should be interpreted with the caveat of cohort-dependent effect sizes.

## Finding 3 — Pathophysiology: endothelial infection drives vasculitis, platelet/coagulation activation, and microthrombi

In MSF patients, in vivo biochemical evidence showed **TXA2-dependent platelet activation** (urinary 11-dehydro-TXB2), **thrombin generation** (prothrombin fragment 1+2), and **endothelial dysfunction** (plasma endothelin-1) ([PMID: 8584998](#)). In *R. conorii*-infected mice, rickettsiae localize to the endothelium and selectively modulate antioxidant enzymes (GPx, GR, G6PD, SOD); the antioxidant alpha-lipoic acid was protective, implicating **oxidative injury** ([PMID: 15120155](#)). Severe ISF can progress to **purpura fulminans and DIC** ([PMID: 29664383](#)).

*"Our results provide biochemical evidence for the occurrence of TXA2-dependent platelet activation and thrombin generation in vivo, together with endothelial dysfunction." — [PMID: 8584998](#)*

## Finding 4 — Clinical phenotype: fever and rash near-universal; eschar frequently ABSENT

In **70 Israeli children**, fever occurred in 100%, rash 98.5%, myalgia 54%, vomiting 40%; thrombocytopenia 75%, hyponatremia 62.5%; 16% hospitalized; 1 death ([PMID: 2710586](#)). Fatal Israeli pediatric cases showed **irreversible shock, encephalopathy, renal failure, and bleeding, with death within 24 h**, often with **no eschar and no tick-bite history** ([PMID: 8286624](#)). A Palestinian pediatric cohort (n=18) reported fever and rash 100%, eschar rare (6%), GI 44%, neurological 28%, arthralgia/myalgia 33%, thrombocytopenia 56%, and universally good outcomes with prompt doxycycline ([PMID: 41795239](#)).

*"the major clinical features were fever (100%), skin rash (98.5%), myalgia (54%) and vomiting (40%). Thrombocytopenia (75%) and hyponatremia (62.5%) were common" — [PMID: 2710586](#)*

*"None of the patients had a history of tick bite, and no tache noire was noted. One child presented without rash" — [PMID: 8286624](#)*

**Documented complications:** hemophagocytic lymphohistiocytosis/HLH ([PMID: 35169573](#), [PMID: 40693676](#)), purpura fulminans ([PMID: 29664383](#), [PMID: 40580427](#)), Guillain-Barré syndrome ([PMID: 41768953](#)), and multi-organ failure in children ([PMID: 26905298](#)).

**Suggested HPO terms:** HP:0001945 (Fever), HP:0000988 (Skin rash), HP:0003326 (Myalgia), HP:0002013 (Vomiting), HP:0001873 (Thrombocytopenia), HP:0002902 (Hyponatremia), HP:0002240 (Hepatomegaly), HP:0001298 (Encephalopathy), HP:0001919 (Acute kidney injury), HP:0031137 (Abnormal circulating coagulation protein concentration), HP:0002624 (vasculitis-related manifestations).

## Finding 5 — Epidemiology: summer, dog-associated, Mediterranean basin with expanding range

The ISF strain is documented in **Israel, Italy (Sicily, Sardinia), and Portugal** ([PMID: 28408252](#)), and *R. conorii* subsp. *israelensis* has been detected in ticks/patients in **Iran** ([PMID: 38254000](#), [PMID: 35365079](#)), **Uganda** ([PMID: 37498943](#)), **Ghana (first record)** ([PMID: 41958159](#)), and ticks in **Israel** ([PMID: 35816829](#)). Seasonality is **late spring/summer**, linked to *Rhipicephalus sanguineus* activity and climate variability/low precipitation ([PMID: 17114701](#)). Dogs serve as a reservoir: **38.5% of 400 healthy Portuguese dogs were seropositive**, and the ISF strain was detected in dog blood ([PMID: 21771547](#)).

*"R. conorii subsp. israelensis, which belongs to the R. conorii complex, is the agent of Israeli spotted fever (ISF); apart from Israel, it has also been found in Italy (Sicily and Sardinia) and in different regions of Portugal."* — [PMID: 28408252](#)

## Finding 6 — Molecular mechanism: Sca-family autotransporters mediate adhesion, invasion, and actin-based motility

Surface cell antigen (Sca) autotransporters—**Sca0/OmpA**, **Sca5/OmpB**, and **Sca2**—mediate host-cell adhesion, invasion, and intracellular motility. **Sca2** (~1800-aa monomeric autotransporter) is sufficient to mediate adherence and invasion of human endothelial cells and drives assembly of **long, unbranched actin tails** for intracellular movement; it is the **only known functional mimic of eukaryotic formins** ([PMID: 22612237](#)). Cryo-EM shows Sca2 forms a formin FH2-dimer-like "doughnut" encircling two actin subunits ([PMID: 37028467](#)).

*"A rickettsial autotransporter from Rickettsia conorii, Sca2, has been shown to be sufficient to mediate both adherence and invasion of human endothelial cells and to participate in intracellular actin-based motility."* — [PMID: 22612237](#)

*"Sca2... is the only known functional mimic of eukaryotic formins"* — [PMID: 37028467](#)

**Suggested GO terms:** GO:0007155 (cell adhesion), GO:0070358 (actin polymerization-dependent cell motility), GO:0030036 (actin cytoskeleton organization).

## Finding 7 — Mechanism of vascular leak: VE-cadherin phosphorylation, nitric oxide, and ROS drive hyperpermeability

SFG rickettsial infection of microvascular endothelial cells activates **tyrosine phosphorylation of VE-cadherin** (peak ~72 h post-infection), reducing VE-cadherin junctional interactions and increasing permeability ([PMID: 22720111](#)). Increased permeability is partly due to intracellular rickettsiae and partly to host pro-inflammatory defenses, with dissociation of endothelial adherens junctions ([PMID: 17957455](#)); **nitric oxide** from infected ECs both limits rickettsial proliferation and alters barrier integrity ([PMID: 16481520](#)). The full pathogenetic sequence—dermal entry → hematogenous spread to endothelium (esp. brain and lungs) → increased permeability, edema, and immunity via NK cells, IFN- $\gamma$ , TNF- $\alpha$ , RANTES, antibodies, and CTLs—is defined in the SFG rickettsioses ([PMID: 12860594](#)).

*"infection of *R. montanensis* significantly activated tyrosine phosphorylation of VE-cadherin beginning at 48 hr and reaching a peak at 72 hr p.i." — PMID: 22720111*

*"The pathogenetic sequence includes rickettsial entry into the dermis, hematogenous dissemination to vascular endothelial cells (most critically in brain and lungs), increased vascular permeability, edema, and immunity mediated by NK cells, IFN-gamma, TNF-alpha, RANTES, antibodies, and cytotoxic T lymphocytes." — PMID: 12860594*

## Finding 8 — Anatomical structures affected: vascular endothelium is the primary target; skin, brain, lung, kidney, liver, GI tract involved

The **microvascular/vascular endothelial cell** is the primary target; disseminated endothelial infection produces increased permeability and edema most critically in **brain and lungs** (PMID: 12860594, PMID: 22720111). Documented organ involvement in ISF/MSF spans skin (rash, eschar, edema), CNS (encephalopathy, meningoencephalitis, Guillain-Barré), lungs (tachypnea, non-cardiogenic pulmonary edema), kidney (azotemia, acute tubular necrosis), liver (hepatomegaly, transaminitis, hyperbilirubinemia), GI tract, spleen, and the hematologic system (thrombocytopenia, coagulopathy, DIC, purpura fulminans) (PMID: 18582199, PMID: 8286624, PMID: 29664383).

*"hematogenous dissemination to vascular endothelial cells (most critically in brain and lungs), increased vascular permeability, edema" — PMID: 12860594*

**Suggested UBERON/CL/GO terms:** UBERON:0001981 (blood vessel), UBERON:0001986 (endothelium), CL:0000115 (endothelial cell), CL:0002139 (endothelial cell of vascular tree); UBERON:0002097 (skin), UBERON:0000955 (brain), UBERON:0002048 (lung), UBERON:0002113 (kidney), UBERON:0002107 (liver); GO:0005912 (adherens junction), GO:0005923 (bicellular tight junction), GO:0005911 (cell-cell junction).

## Finding 9 — Prevention is non-pharmacologic: no vaccine; tick-bite avoidance and vector control

No licensed human vaccine exists for *R. conorii*/SFG rickettsiae; prevention depends on **avoiding tick bites and controlling the brown dog tick**. Integrated vector control (surveillance, residual acaricide spraying, dog tick collars, public outreach) substantially reduced *Rhipicephalus sanguineus* populations and rickettsial risk in a California program

(PMID: 41146265). Effective canine ectoparasiticides include **isoxazolines (e.g., lotilaner)** and combination endectocides (PMID: 42045933); pyrethroid acaricides target the tick voltage-gated sodium channel but **resistance is emerging** (PMID: 41272771). Early empiric doxycycline is the key **secondary-prevention** measure against fatal outcomes.

*"the integrated intervention substantially reduced tick populations at the affected site. Both adult and immature stages of Rh. sanguineus s.l. declined following sequential treatments."*  
— PMID: 41146265

### Finding 10 — ISF is a zoonosis: dogs are reservoir/sentinel hosts

*R. conorii* subsp. *israelensis* DNA has been detected in *Rhipicephalus sanguineus* dog ticks in **Ghana** (first record; 3.95% of pools; PMID: 41958159), **Uganda** (PMID: 37498943), and **Israel** (PMID: 35816829); the ISF strain was detected in Portuguese dog blood with 38.5% seropositivity (PMID: 21771547). The tick maintains the agent transovarially/transstadially, and the ISF strain is better tolerated by the tick than Malish (PMID: 19421877). Naturally infected species include domestic dogs (*Canis lupus familiaris*, NCBI Taxon 9615) and the tick vector (*Rhipicephalus sanguineus*, NCBI Taxon 34632).

*"This study reports the first molecular detection of R. conorii subsp. israelensis in Ghana."* — PMID: 41958159

### Finding 11 — Treatment: doxycycline is drug of choice; josamycin/clarithromycin are alternatives

A randomized trial of **1-day doxycycline vs 5-day josamycin in 59 MSF patients** showed all recovered uneventfully with no significant difference; single-day doxycycline is effective, easy, and inexpensive (PMID: 2193627). An RCT of **clarithromycin vs doxycycline/josamycin (n=40, incl. 13 children <14 y)** found no significant difference in time to defervescence (2.67 vs 2.22 d) or symptom resolution, with no adverse reactions or relapses (PMID: 26711765). Josamycin is favored first-choice in **children and pregnant women** (PMID: 1884779). Chloramphenicol is effective but carries bone-marrow toxicity, so tetracyclines are preferred.

*"One-day doxycycline therapy is an effective, easy, and inexpensive treatment. Josamycin is a useful therapeutic alternative that may be particularly convenient for pregnant women and patients with a history of allergy to tetracyclines."* — [PMID: 2193627](#)

*"clarithromycin is a good alternative to doxycycline or josamycin in the treatment of MSF."* — [PMID: 26711765](#)

**Suggested NCIT/CHEBI terms:** NCIT:C692 (Doxycycline), NCIT:C1032 (Clarithromycin), NCIT:C376 (Chloramphenicol); CHEBI:50845 (doxycycline), CHEBI:3732 (clarithromycin).

## Finding 12 — Temporal course & prognosis: acute, self-limited after ~6-day incubation; excellent with early treatment

Onset is **acute after a short (~5–7 day) incubation**; severe ISF can progress to neurologic and multi-organ failure within days, with recovery achievable if appropriate antibiotics start early ([PMID: 14964019](#)). Fatal pediatric ISF progressed to shock, encephalopathy, and renal failure with death within 24 h of admission ([PMID: 8286624](#)). In the strain-identified Portuguese cohort, **29/140 (21%) adults died**, fatality significantly higher for ISF ([PMID: 18582199](#)). **Prognostic factors for death:** ISF strain, older age, alcoholism, delayed treatment, G6PD deficiency, petechial/purpuric rash, obtundation, coagulopathy, azotemia, hepatic dysfunction.

*"Clinical disease was characterized by irreversible shock, encephalopathy, renal failure, bleeding tendency, and death within 24 hours of admission."* — [PMID: 8286624](#)

## Finding 13 — Diagnosis: IFA serology (gold standard) plus PCR/sequencing of skin biopsy targeting *gltA*/*ompA*

In 128 confirmed MSF patients, **IFA** showed seroconversion or  $\geq 4$ -fold titre rise in 97 (77%) and a single high titre in 16 (12.7%); **skin-biopsy PCR** was positive in 77/106 (72.6%), with similar yield from eschar (73%) and maculopapular rash (70%) specimens ([PMID: 23168048](#)). Real-time PCR of skin biopsy plus **DNA sequencing of *gltA* and *ompA*** enables species/subspecies identification ([PMID: 40608626](#)), and ISF has been specifically confirmed by real-time PCR + IFA in case reports ([PMID: 38254000](#)).

*"Using IFA, seroconversion or a fourfold or greater rise in titre was observed in 97 (77%) patients, whereas a single high titre was demonstrated in 16 (12.7%) patients. According to PCR analysis, 77 (72.6%) of 106 biopsy samples showed positive results." — PMID: 23168048*

**Common laboratory abnormalities (LOINC-relevant):** thrombocytopenia, hyponatremia, elevated transaminases, hyperbilirubinemia, elevated CRP, azotemia, elevated creatine kinase. **Differential diagnosis:** other SFG rickettsioses (African tick-bite fever — more numerous eschars, PMID: 41244315), meningococemia, leptospirosis, viral exanthems, and other causes of fever + rash. There is **no genetic testing, newborn screening, or omics-based diagnostic** relevant to this infectious disease.

### Finding 14 — Animal models: susceptible C3H/HeN(J) vs resistant C57BL/6 mice; immunity via NK cells, IFN- $\gamma$ , CD4 Th1

**C3H/HeN and C3H/HeJ mice** are susceptible and develop lethal disseminated infection with endothelial targeting; **C57BL/6 mice are resistant** (PMID: 17403875). NK-cell depletion enhances susceptibility, and early control is **IFN- $\gamma$ /IL-12 dependent** (PMID: 11504408). Susceptible C3H mice show delayed CD4<sup>+</sup> Th1/Th2 responses and higher Foxp3<sup>+</sup> Tregs. The ISF (ISTT) strain differs from Malish in tick biology: it caused less tick mortality and higher tick infection prevalence (35–66% vs <5%) in *Rhipicephalus sanguineus* (PMID: 19421877). Tick saliva modulates host inflammation (IL-1 $\beta$ , NF- $\kappa$ B) during transmission (PMID: 26011701).

*"depletion of NK cell activity with antibody to asialo GM1 enhanced the susceptibility of C3H/HeN mice to infection with R. conorii" — PMID: 11504408*

*"exposure to ISTT strain had lesser effect on tick survival and resulted in 35-66% prevalence of infection" — PMID: 19421877*

**Model organisms:** *Mus musculus* (NCBI Taxon 10090) — inbred C3H/HeN, C3H/HeJ (susceptible), C57BL/6 (resistant); primary human microvascular/umbilical vein endothelial cells for in vitro work. **Recapitulation:** disseminated endothelial infection, vasculitis, and lethal multi-organ disease are well reproduced in susceptible mice; **limitations** include that eschar/tache noire formation and the human tick-bite transmission route are incompletely modeled without live tick infestation.

## Finding 15 — Israeli national epidemiology: predominant SFG rickettsiosis, ~30% case-fatality, coastal clustering

A nationwide Israeli study (2010–2019) identified **42 hospitalized SFG rickettsiosis cases (36 autochthonous)**; the *R. conorii* Israeli tick typhus strain was most prevalent (**33/42, 79%**), required intensive care in 52%, and had a **30% fatality rate**. History of tick bite was present in only 5%; eschar in 12%; leukocytosis more common than leukopenia; **72% resided along the Mediterranean shoreline** (PMID: 34286684). A seroepidemiologic survey of two Israeli villages found **69/85 (81%) dogs vs 14/136 (10%) humans** anti-*R. conorii* seropositive, establishing canine serology as a sensitive sentinel for human exposure (PMID: 17620644). ISF in Sicily can be traced back to 1987–1991, with 3/5 patients severe and 1 death (PMID: 16333093).

*"The most prevalent species was the Rickettsia conorii Israeli tick typhus strain (n = 33, 79%); infection with this species necessitated intensive care for 52% of patients and was associated with a 30% fatality rate. A history of tick bite was rare, found for only 5% of patients; eschar was found in 12%" — PMID: 34286684*

*"Sixty-nine of 85 (81%) canine sera and 14 of 136 (10%) of human sera had anti-R. conorii antibodies." — PMID: 17620644*

## Finding 16 — Genetic/molecular etiology: an infectious, not Mendelian, disease

ISF is an **infectious**, not a Mendelian genetic, disease; there are **no human causal genes, pathogenic variants, inheritance patterns, penetrance/expressivity considerations, founder effects, carrier frequencies, or chromosomal abnormalities** to report. Host genetic modifiers of severity are plausible but under-characterized: **G6PD deficiency** is repeatedly cited as a risk factor for severe/fatal MSF, consistent with the oxidative-injury mechanism and the selective modulation of G6PD observed in infected tissues (PMID: 15120155). In animal models, host background (C3H vs C57BL/6) strongly determines susceptibility, implying host genetic control of outcome (PMID: 17403875). The relevant "genome" for causal variants is that of the **pathogen** (*sca0/ompA*, *sca5/ompB*, *sca2*, *sca4*, *gltA*, *16S rRNA*), which defines strain identity and virulence.

## Mechanistic Model / Interpretation

### Ordered causal chain (initiating lesion → clinical manifestation)

1. Infected *Rhipicephalus sanguineus* tick bite inoculates *R. conorii* subsp. *israelensis* into the dermis —leads to→
2. Sca0/OmpA + Sca5/OmpB mediate adhesion to vascular endothelial cells —results in→
3. Receptor-mediated invasion of endothelial cells (Sca2 sufficient for entry) —leads to→
4. Intracellular replication + Sca2 formin-mimic actin-based motility —drives→
5. Cell-to-cell spread and hematogenous dissemination to systemic microvasculature (brain and lungs most critical) —results in→
6. Endothelial injury:
  - └ VE-cadherin tyrosine phosphorylation → adherens-junction dissociation
  - └ Nitric oxide + reactive oxygen species (ROS) → peroxidative membrane damage
  - └ Pro-inflammatory host response (NK cells, IFN- $\gamma$ , TNF- $\alpha$ , RANTES, CTLs)
  - all converge on→
7. Increased microvascular permeability → interstitial edema (cerebral, pulmonary) + small-vessel vasculitis —in parallel with→
8. Platelet activation (TXA2) + thrombin generation → procoagulant state, microthrombi —branches→
  - └ Usual course: self-limited vasculitis → fever + maculopapular rash  
→ RESOLUTION with early doxycycline
  - └ Severe/ISF-strain course: DIC, purpura fulminans, HLH, shock, encephalopathy, renal failure, multi-organ failure → DEATH (20–30%)

Steps 1–5 and the endothelial tropism (step 6) are **experimentally demonstrated** (in vitro human ECs, mouse models, patient biopsies). The relative contribution of intrinsic rickettsial cytotoxicity vs host immunopathology to permeability (step 7) is **partly inferred**; both are supported ([PMID: 17957455](#)). Why the ISF strain is more virulent than Malish at the molecular level is **not yet mechanistically resolved**—it is an empirical epidemiologic/clinical observation ([PMID: 18582199](#), [PMID: 34286684](#)).

## Upstream vs downstream summary

| Level                    | Mechanism                                                | Evidence source                                                                              | GO/CL suggestion                                                        |
|--------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| Upstream<br>(trigger)    | Tick inoculation; Sca-mediated adhesion/<br>invasion     | In vitro, cryo-EM<br>( <a href="#">PMID: 22612237</a> ,<br><a href="#">PMID: 37028467</a> )  | GO:0007155 cell adhesion;<br>CL:0000115 endothelial cell                |
| Middle                   | Actin-based motility;<br>hematogenous spread             | In vitro, mouse<br>( <a href="#">PMID: 15120155</a> )                                        | GO:0070358 actin<br>polymerization-based<br>movement                    |
| Downstream<br>(effector) | VE-cadherin<br>phosphorylation, NO/<br>ROS, permeability | In vitro human ECs<br>( <a href="#">PMID: 22720111</a> ,<br><a href="#">PMID: 16481520</a> ) | GO:0005923 tight junction;<br>GO:0034220 ion<br>transmembrane transport |
| Terminal<br>(clinical)   | Vasculitis, edema, DIC,<br>multi-organ failure           | Patient cohorts<br>( <a href="#">PMID: 18582199</a> ,<br><a href="#">PMID: 34286684</a> )    | HP:0002624 vasculitis;<br>HP:0001928 coagulopathy                       |

## Strain comparison table

| Feature                   | Malish ( <i>R. conorii</i> subsp. <i>conorii</i> ) | ISF ( <i>R. conorii</i> subsp. <i>israelensis</i> )                                                  |
|---------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------|
| MLST genotype             | Malish type                                        | ISFR type ( <a href="#">PMID: 15766388</a> )                                                         |
| Eschar (tache noire)      | ~49% (Portugal)                                    | ~39% (Portugal); ~5–12% (Israel) ( <a href="#">PMID: 16481514</a> , <a href="#">PMID: 34286684</a> ) |
| Virulence / fatality      | Lower                                              | Higher; fatal outcome significantly more likely<br>( <a href="#">PMID: 18582199</a> )                |
| Tick infection prevalence | <5%                                                | 35–66%; better tolerated by tick ( <a href="#">PMID: 19421877</a> )                                  |

## Evidence Base

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| PMID                     | Title (abbrev.)                                 | Contribution                                                     |
|--------------------------|-------------------------------------------------|------------------------------------------------------------------|
| <a href="#">15766388</a> | MLST subspecies of <i>R. conorii</i>            | Defines ISF as subsp. <i>israelensis</i> ; taxonomy backbone     |
| <a href="#">18582199</a> | Fatal <i>R. conorii</i> risk factors (Portugal) | Core evidence: ISF strain virulence, 21% fatality, alcoholism    |
| <a href="#">34286684</a> | SFG rickettsioses in Israel 2010–2019           | National epidemiology; 79% ISF, 30% fatality, coastal clustering |
| <a href="#">12860594</a> | Pathogenic mechanisms of <i>Rickettsia</i>      | Full pathogenetic causal chain and immune mediators              |
| <a href="#">22720111</a> | Rickettsiae + VE-cadherin (AFM)                 | Molecular basis of endothelial hyperpermeability                 |
| <a href="#">22612237</a> | Sca2 domains                                    | Sca2 adhesion/invasion/motility                                  |
| <a href="#">37028467</a> | Cryo-EM of Sca2                                 | Formin-like core structure                                       |
| <a href="#">8584998</a>  | Coagulation/platelet activation in MSF          | In vivo procoagulant state, endothelial dysfunction              |
| <a href="#">2710586</a>  | Spotted fever in Israeli children               | Symptom/lab frequencies                                          |
| <a href="#">8286624</a>  | Fatal ISF in children                           | Absence of eschar/tick history; rapidly fatal course             |
| <a href="#">41795239</a> | Palestinian pediatric cohort                    | Modern pediatric phenotype; benign with early Rx                 |
| <a href="#">23168048</a> | MSF in Turkey                                   | Diagnostic yield of IFA + biopsy PCR                             |
| <a href="#">40608626</a> | Rickettsiosis case series                       | gltA/ompA sequencing for species ID                              |
| <a href="#">2193627</a>  | RCT doxycycline vs josamycin                    | First-line therapy evidence                                      |
| <a href="#">26711765</a> | RCT clarithromycin                              | Macrolide alternative for children                               |
| <a href="#">21771547</a> | <i>R. conorii</i> in Portuguese dogs            | Reservoir/sentinel role of dogs                                  |
| <a href="#">17620644</a> | Seroepidemiology, Israeli villages              | Dogs 81% vs humans 10% seropositive                              |
| <a href="#">19421877</a> | ISTT vs Malish in ticks                         | Strain difference in vector biology                              |
| <a href="#">17403875</a> | Dendritic cells & susceptibility                | C3H susceptible vs C57BL/6 resistant model                       |
| <a href="#">11504408</a> | NK/IFN- $\gamma$ early immunity                 | Protective immune mechanism                                      |

| PMID                     | Title (abbrev.)                        | Contribution                                                    |
|--------------------------|----------------------------------------|-----------------------------------------------------------------|
| <a href="#">28408252</a> | ISF in Sicily                          | Geographic distribution; disease identity                       |
| <a href="#">16481514</a> | Eschar in ISF patients                 | Nuance: no statistical eschar/severity difference in one cohort |
| <a href="#">41146265</a> | Integrated vector control (California) | Prevention via <i>Rh. sanguineus</i> control                    |
| <a href="#">41958159</a> | ISF agent in Ghana ticks               | Expanding geographic range; zoonotic cycle                      |

**Challenging / nuancing evidence:** [PMID: 16481514](#) found no *statistically significant* difference in eschar frequency or fatality between Malish and ISF in a 94-patient Portuguese cohort, tempering the "greater severity" narrative that is otherwise strongly supported by the larger prospective cohort ([PMID: 18582199](#)) and Israeli national data ([PMID: 34286684](#)). The most likely reconciliation is that strain virulence interacts with host factors (age, alcoholism, G6PD status) and treatment timing, so cohort composition strongly influences the observed effect size.

## Limitations and Knowledge Gaps

- 1. Molecular basis of ISF hypervirulence is unresolved.** Why *R. conorii* subsp. *israelensis* causes higher fatality than Malish is documented epidemiologically but not explained at the genomic/proteomic level. No comparative virulence-factor study isolating the responsible loci was identified.
- 2. No dedicated ISF pathophysiology datasets.** Most mechanistic findings derive from related SFG rickettsiae (*R. rickettsii*, *R. montanensis*) and MSF broadly, not ISF-specific experiments. Extrapolation is reasonable but formally an inference.
- 3. Human host-genetic modifiers are under-characterized.** G6PD deficiency is repeatedly cited but lacks a large, controlled association study for ISF specifically.
- 4. Epidemiology is likely under-ascertained.** Rare tick-bite history and frequent absence of eschar cause diagnostic delay and underdiagnosis; true incidence/prevalence figures (cases per 100,000) are not well established outside hospitalized cohorts.
- 5. No quality-of-life instruments (EQ-5D, SF-36) have been applied** to ISF survivors; long-term sequelae (e.g., post-GBS, post-HLH) are described only in case reports.

6. **Publication/selection bias toward severe cases.** Case reports over-represent complications (purpura fulminans, HLH), while population data ([PMID: 41795239](#)) suggest most pediatric disease is benign when treated early.
  7. **No omics (transcriptomics/proteomics/metabolomics) profiling specific to ISF** was found; molecular-profiling sections are supported only by model-organism antioxidant-enzyme and lncRNA studies.
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## Proposed Follow-up Experiments / Actions

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1. **Comparative genomics/transcriptomics of ISF vs Malish strains** in matched human microvascular endothelial cell infections to pinpoint virulence determinants (candidate loci: *sca* family, *rickA*, toxin-antitoxin modules). Read out permeability (TEER), VE-cadherin phosphorylation, and cytokine induction.
  2. **Case-control study of G6PD deficiency and other host modifiers** (HLA, cytokine polymorphisms) in strain-confirmed ISF cohorts to quantify the contribution of host genetics to fatality.
  3. **Prospective natural-history and QoL study** (EQ-5D/PROMIS) of ISF survivors, including neurological and hematologic sequelae, to fill the outcomes/prognosis gap.
  4. **Enhanced One-Health surveillance** integrating canine serology (a sensitive sentinel; [PMID: 17620644](#)) and tick molecular screening across the expanding geographic range (Iran, sub-Saharan Africa) to define true incidence and detect range expansion driven by *Rhipicephalus sanguineus* spread and climate change.
  5. **Evaluate acaricide-resistance management** for *Rh. sanguineus* (isoxazoline rotation) given emerging VGSC pyrethroid-resistance mutations ([PMID: 41272771](#)), and model integrated vector-control programs for ISF-endemic coastal communities.
  6. **Clinician decision-support/awareness intervention:** because tick bite and eschar are frequently absent, promote empiric doxycycline for summer fever+rash in endemic/coastal areas to shorten the treatment delay that drives mortality.
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## Consensus Answer

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Israeli Tick Typhus (Israeli spotted fever; MONDO:0000230) is a tick-borne spotted-fever-group rickettsiosis caused by the obligate intracellular bacterium *Rickettsia conorii* subsp. *israelensis*, transmitted by the brown dog tick *Rhipicephalus sanguineus* with dogs as the amplifying reservoir. It infects vascular endothelial cells to produce a disseminated small-vessel vasculitis presenting as fever and maculopapular rash (with the tache noire/eschar frequently absent), and the more virulent ISF strain can progress to severe multi-organ disease with case-fatality reaching ~20–30% in hospitalized/severe cases. Prompt empiric doxycycline is the first-line treatment and is highly effective when started early.

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